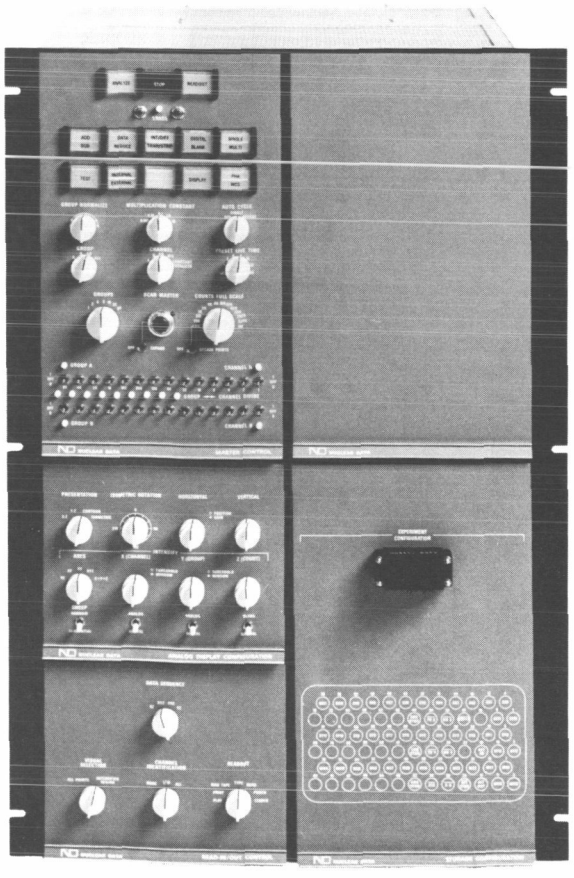
TECHNICAL SPECIFICATIONS

NUCLEAR DATA INC.
100 West Golf Road
Palatine, Illinois 60067
Phone A.C. 312 • 529-4600



SERIES 3300 DATA HANDLING MAIN FRAME AND STORAGE CONFIGURATION

The nucleus of the basic Series 3300 Analyzer System is a modular data handling unit. It consists of a main signal distribution frame and modules associated with data handling functions. The signal distribution frame, which provides the mechanical and electrical facilities for acceptance of the modules, also incorporates digital circuitry required for data processing and related operations. Construction of the signal distribution frame is such that it can be housed in a standard relay rack or other similar enclosure.

Modules included in the basic Series 3300 Data Handling Unit are Master Control, Read-In/Out Control, Analog Display Configuration, and Storage Configuration. The Storage Configuration Module, which determines the storage format of data supplied by the System Data Acquisition Unit, can be replaced by other input data conditioning modules such as "Digital Band Selector" and "Region of Interest". A vacant module housing is also included with the basic Series 3300 Data Handling Unit. Its position can be occupied by a module such as a computer interface or an automatic program sequencer, if expansion of system capabilities is desired.

DATA HANDLING MAIN FRAME

Logic Levels

Standard logic levels of plus six (+6) volts for a logical "one" or true condition, and zero (0) volts for a logical "zero" or false condition are used throughout to facilitate system testing and to simplify interfacing, if required.

Address Register

The Address Register can retain sixteen bits of binary information. It can accept up to fourteen binary bits of information by means of parallel transfer, and provides parallel access to each bit for transfer out purposes. Up to fourteen of the least significant bits are used to locate storage locations in the System Memory. The two most significant bits are used exclusively for operations associated with the system display function.

In addition, the Address Register will scale at rates of up to one megacycle per second (1 MHz) and will operate as a shift register for system display functions.

Memory Register

The Memory Register can retain up to twenty bits of binary information. It will operate as an add or subtract scaler at rates of at least one megacycle per second (1 MHz). It provides binary information for storage in the System Memory during memory "write" operation, and accepts information of the

same bit content during memory "read" operation. Information can also be entered into the Memory Register by means of parallel transfer, and parallel access is provided for transfer out purposes.

Data Processing Register

The Data Processing Register accommodates up to twenty-four bits of binary or binary-coded decimal information. It will operate as either an add or subtract scaler, and forward or backward shift register. It is used in the modification of stored data during operations such as integration, stripping, and algebraic addition with data read from external sources.

In addition, the Data Processing Register is used in the formatting and binary to binary-coded decimal conversion of data for appropriate readout devices. It is also used in the reverse operation of conversion and formatting of data read in from external sources for memory storage or algebraic addition with previously stored data.

Serial Binary Adder

A Serial Binary Adder circuit is included for use in operations such as integration, stripping, and algebraic addition of read-in data with stored data.

Main Control Circuits

All flip-flops associated with the main operating modes of the system, i.e., Accumulate, Display, Readout, Stop, etc., are included.

Register Buffers

All buffers required for operations with the Address Register, Memory Register, and Data Processing Register are included.

Readout Timing Control

a. Digital Readout and Read-in
A timing control scaler for generation of program pulses during digital readout and read-in operations is included.

b. Magnetic Tape Readout and Read-in
A timing control scaler for generation of program pulses during magnetic tape readout and read-in operations is included.

Digital Readout and Read-in Solenoid Drivers

The solenoid drivers required for operation of appropriate digital readout and read-in devices is included.

Input/Output Connectors

Connectors are provided on a rear apron for cabling to and from the Data Acquisition Section, to and from the System Memory, to and from a Magnetic Tape Control Main Frame, to and from an external programmer, and to and from the System Power Supply. Also provided are connectors for operation of a high-speed parallel printer, paper tape punch, paper tape reader, typewriter, display oscilloscope, and X-Y recorder. Facilities for operation of a high-speed point plotter are also included.

Synchronization Points

A rear apron switch enables selection of oscilloscope synchronization points for system test and checking.

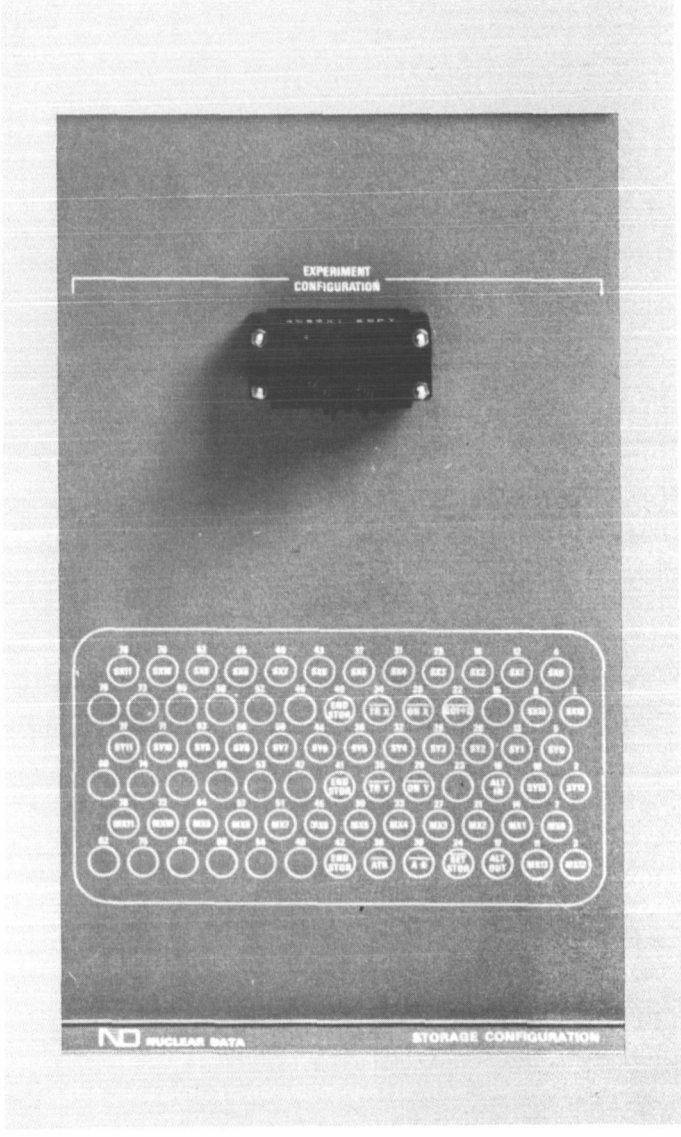
Sign Identification

For digital readout, disabling of non-significant zero suppression for negative number obtained by over-subtracting can be selected by a rear apron switch.

STORAGE CONFIGURATION MODULE

Program Receptacle

A front panel program board receptacle accepts a variety of pre-wired program plugs which establish the storage configuration of the memory. Storage configurations are $2^n \times 2^n$ where 2^n can be any binary multiple, providing the product of the two multiples does not exceed the number of channels in the system memory.



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